

Senior Software Engineer

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SUMMARY

Senior Software Engineer with 11 years of experience designing, building, and delivering **reliable**, **scalable**, and **production-ready** software across backend, mobile, frontend, systems programming, tooling, and real-time application domains. Strong background in backend architecture, data-driven applications, microservices, caching, networking, native integrations, build automation, compiler/toolchain work, and performance-sensitive software. Experienced in solving complex technical problems, improving maintainability, supporting production delivery, and collaborating across engineering teams to build practical and robust systems.

WORK EXPERIENCE

Sharif AI Research Center | Senior Software Engineer Jan 2025 - Present

- Contributing to the design and implementation of a compiled programming language and its supporting compiler/toolchain infrastructure.
- Working on systems-level components involving LLVM integration, linking, library management, allocator-related systems, and runtime behavior.
- Designing and maintaining low-level infrastructure for code generation, build workflows, memory management, and developer tooling.
- Collaborating on complex engineering problems related to language design, runtime reliability, tooling architecture, and maintainable system design.

DeltaClash | Software Engineer May 2023 - Jan 2025

- Developed real-time client-side application systems and interactive features using Haxe and the Heaps engine.
- Designed and implemented a scripting engine in Haxe to support configurable behavior, application logic, and modding-style extensibility.
- Worked on backend programming and server networking design for real-time client/server systems.
- Improved maintainability by separating application logic, scripting behavior, networking responsibilities, and engine-level concerns.

Taawoos Studio | Software Engineer May 2021 - May 2023

- Worked on **Lucid Lenses**, implementing core application systems, interactive logic, and production features.
- Designed optimized graph traversal logic for complex state-based mechanics and application behavior.
- Created a node-based dialogue and flow system for managing structured interactions and branching application states.
- Implemented and maintained production application features with a focus on performance, stability, maintainability, and delivery quality.

BlueBoxGames | Full-Stack Software Developer Sep 2020 - Nov 2020

- Worked across the **client**, **core**, and **server** components of a client/server application.
- Developed backend services, shared application logic, and client-side functionality for a cross-platform architecture.
- Built native extensions, including the **Metrix Native Extension**, to integrate platform-specific functionality.
- Connected native platform capabilities with higher-level application code to support cross-platform builds and production delivery.

Earlier Experience | Independent Software Developer 2014 - 2020

- Built websites and backend applications using Django and Python.
- Developed practical experience across backend development, mobile applications, debugging, low-level programming, application architecture, and production-oriented software delivery.

TECHNICAL SKILLS

Programming Languages

- C, C++, C#, Java, Haxe, TypeScript, JavaScript
- Working knowledge of Go, Rust, OCaml, Ruby, Python, and Lua

Backend and Application Architecture

- REST APIs, GraphQL, service-oriented architecture, backend application design
- Microservices, asynchronous processing, Redis caching, RabbitMQ message queues
- Express.js, Node.js ecosystem, networking fundamentals, concurrency fundamentals
- Knowledge of distributed systems concepts and gRPC for service communication

Databases and Data Storage

- PostgreSQL, MySQL/MariaDB, SQLite
- MongoDB, PouchDB
- Data modeling, query optimization, large data handling, offline-first data synchronization

Mobile and Cross-Platform Development

- Java-based Android application development and React Native application development
- Kotlin-based Android code familiarity, basic Swift and iOS development familiarity
- Java Native Interface development, native module integration, platform integration, and cross-platform production builds

Systems Programming and Tooling

- Data structures, algorithms, compiler and interpreter design
- Lexer, parser, code generation, runtime systems, memory management, and low-level debugging
- LLVM integration, build systems, native libraries, and performance-sensitive programming

DevOps, Build and Delivery

- CI/CD workflows, automated testing, release builds, build automation
- Make, CMake, Git, Linux
- Production build delivery, deployment support, and maintainable development workflows

Graphics, Real-Time Systems and Interactive Applications

- Real-time application architecture, rendering fundamentals, OpenGL, SDL
- Unity, Heaps, OpenFL, Starling
- Native extensions, engine-level programming, performance optimization, and interactive system design

Geospatial and GIS

- GIS fundamentals, geospatial data concepts, coordinate systems, and map-based applications
- Understanding of location-based systems, spatial data processing, and geospatial algorithms

PROJECTS

cashash.c | C

Personal Project

- Developed a generic hash table library in C with support for binary key-value pairs.
- Implemented collision handling using separate chaining and open addressing, including linear probing, quadratic probing, and double hashing.
- Designed APIs for insertion, deletion, lookup, iteration, and internal hash table validation.
- Added unit tests to verify collision handling, deletion, iteration, and correctness of different hash strategies.

Z Programming Language | C/C++

Sharif AI Research Center

- Designed and implemented components of the Z Programming Language and its compiler/toolchain infrastructure.
- Worked on systems-level language features involving parsing, code generation, linking, library management, and runtime behavior.

HTTP Request Parser | C

Personal Project

- Built a fast HTTP request parser in C focused on parsing performance, low-level control, and networking fundamentals.
- Implemented parser logic suitable for systems-level backend infrastructure experiments.

Myna | Java, OpenGL ES

Personal Project

- Built a simple experimental 2D engine in native Java for Android.
- Used native Android features and OpenGL ES for rendering, application behavior, and mobile graphics experimentation.

Dropting | Haxe, Heaps, WebSocket

Personal Project

- Developed the client application using the Heaps engine.
- Built the server-side component in Haxe using Colyseus WebSocket.
- Designed a client/server structure for real-time interactive application behavior.

Maze Generator Logic Library | C

Personal Project

- Built a reusable maze generation library in C with support for passing generation algorithms as functions.
- Used libatopng to generate visual output from generated maze data.

EDUCATION

Payame Noor University | Master's in Computer Science and Engineering

Jan 2019 - May 2023

Allameh Tabataba'i University | Bachelor's in Tourism Management

Sep 2014 - May 2018